

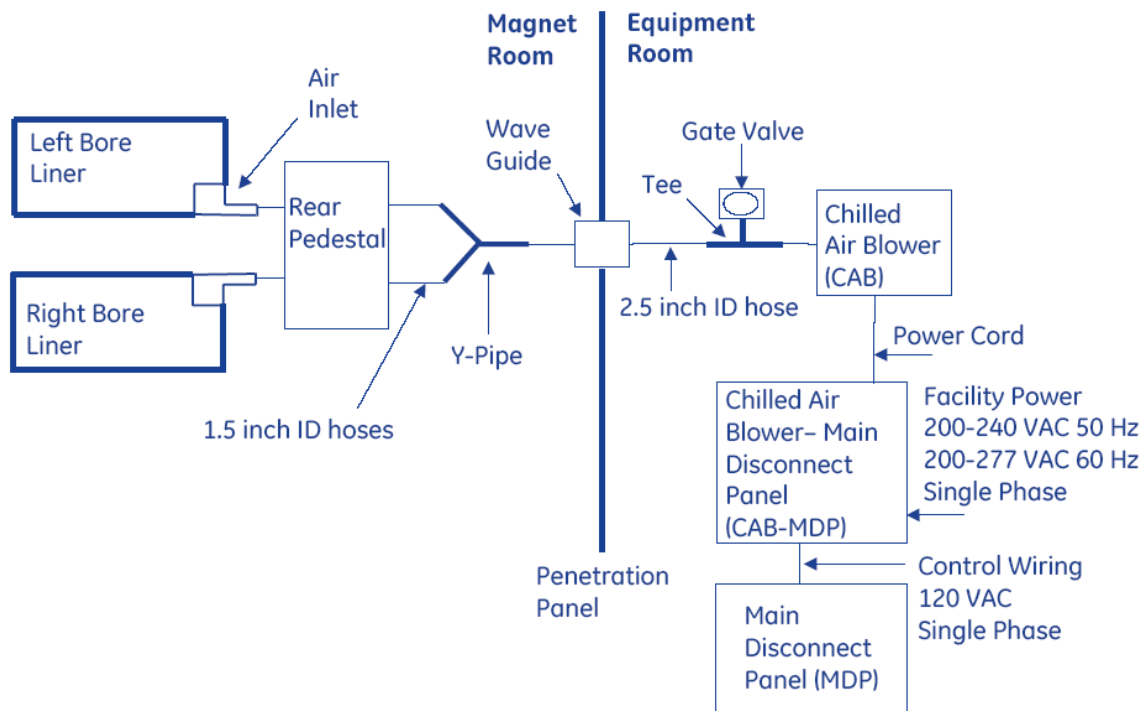
# Integrated Patient Comfort Module (IPCM)

## Option Installation

### 1 Installation Overview

The GE IPCM Option consists of hardware and software that can be used in combination with fMRI scanning. This document describes the hardware and software installation for this option. The hardware consists of the Chilled Air Blower (CAB), the CAB Main Disconnect Panel (CAB-MDP), Bore Liner hardware, Software option key or update CD, and the miscellaneous hardware to mount and connect the system components together. See Illustration 1 for the overview diagram.

**Illustration 1:** Overview Diagram



## 1.1 Options Installed by This Manual

This manual describes installation of the following GEHC MR catalog options:

**Table 1:** IPCM MR Catalog Options

| MR Catalog Number | Description                          |
|-------------------|--------------------------------------|
| M3033LG           | 3T Excite II systems (G3 software)   |
| M3033LH           | 3T Excite HD systems (12x software)  |
| M3033LJ           | 3T Excite HDx systems (14x software) |

## 1.2 Site Readiness

### Hardware Requirements:

This option requires pre-installation work to be done ahead of the complete installation. This work includes placing the Chilled Air Blower (CAB), and mounting and supplying power to the Chilled Air Blower Main Disconnect Panel (CAB-MDP). See the associated PIM manual for the specific requirements.

### Electrical work that needs to be done by the customer.

1. Provide Electrical Source to the CAB-MDP  
Specification: 200-240 V, Single Phase 50 Hz  
200-277 V, Single Phase 60 Hz
2. Provide control circuit between the MDP and the CAB-MDP.  
Specification: 110V, Single Phase.  
Typical #12 gauge wire.

### Software Requirements:

This option requires a certain software revision before the IPCM option key can be installed. The minimum requirements are listed below:

**3.0T HD Systems: 12.0 M5**

**3.0T HDx Systems: 14.0 M3**

## 1.3 Shipping List

Table 2 lists the contents of each catalog.

**Table 2:** IPCM MR Catalog Options

| Item      | Qty | Description                                           | M3033LG<br>Part<br>Number | M3033LH<br>Part<br>Number | M3033LJ<br>Part<br>Number |
|-----------|-----|-------------------------------------------------------|---------------------------|---------------------------|---------------------------|
| <b>1</b>  |     | Bore Liner Hardware                                   | 5126928                   | 5126928                   | 5126928                   |
|           | 1   | Y Pipe Bonded Assembly For Boreliner                  |                           |                           |                           |
|           | 2   | Corrugated Pvc Air Hose, 1.5 Inch Id, 8 FT (2.4M)     |                           |                           |                           |
|           | 1   | 2 Inch Pvc Threaded Tee                               |                           |                           |                           |
|           | 1   | Pvc Air Duct 80 FT (24.4 M)                           |                           |                           |                           |
|           | 2   | 2 Inch Pvc Nipple, Npt                                |                           |                           |                           |
|           | 1   | 2 Inch Pvc Gate Valve, 2 Inch Npt Female              |                           |                           |                           |
|           | 2   | Pvc Hose Adapter Barb X Male For 2 Inch Hose Id       |                           |                           |                           |
|           | 10  | 1.5625 Inch To 2.5 Inch Sst Worm Drive Hose Clamp     |                           |                           |                           |
|           | 4   | Set Screw                                             |                           |                           |                           |
|           | 2   | Foam Insert For Bore Liner Bridge                     |                           |                           |                           |
|           | 2   | 1 Inch X 6 Inch Dual Lock Air Inlet Piece             |                           |                           |                           |
|           | 4   | Nut Hexagon M10 A2-70                                 |                           |                           |                           |
|           | 1   | Chilled Air Blower Instruction Sheet                  |                           |                           |                           |
|           | 1   | IPCM Installation Manual                              | 5165538                   | 5165538                   | 5165538                   |
| <b>2</b>  | 1   | Chilled Air Blower (CAB)                              | 5118421                   | 5118421                   | 5118421                   |
| <b>3</b>  |     | Thermal Bore Liner Assembly Parts                     | 5132472                   | 5132472                   | 5132472-2                 |
|           | 1   | Thermal Bore Liner Assembly Right                     |                           |                           |                           |
|           | 1   | Thermal Bore Liner Assembly Left                      |                           |                           |                           |
|           | 1   | Air Inlet Assembly Right                              |                           |                           |                           |
|           | 1   | Air Inlet Assembly Left                               |                           |                           |                           |
|           | 1   | Bore Liner Attachment Hardware                        |                           |                           |                           |
| <b>4</b>  | 1   | Auxiliary Power Panel (CAB-MDP)                       | 5133079                   | 5133079                   | 5133079                   |
| <b>5</b>  | 1   | Modified Pen Panel                                    | 5133889                   | 5133889                   | 5133889                   |
| <b>6a</b> | 1   | G3 and HD Bridge with Notched Sides (color off white) | 2386348-2                 | 2386348-2                 |                           |
| <b>6b</b> | 1   | Hdx Bridge with Notched Sides (color white)           |                           |                           | 5150910-2                 |
| <b>7a</b> | 1   | 12x Software option key                               |                           | 5165551                   |                           |
| <b>7b</b> | 1   | 14x Software option key                               |                           |                           | 5164641                   |
| <b>8</b>  | 1   | HD Rear Split Bridge                                  | 2386347                   |                           |                           |
| <b>9</b>  | 1   | Split Bridge Chain Bracket                            | 2386349                   |                           |                           |

## 1.4 Tools Required

| Description                            | Part Number   | Quantity            |
|----------------------------------------|---------------|---------------------|
| Lockout/Tagout Multiple Locking Device | 46-194427P313 | 2                   |
| Brass Master Padlock                   | 46-194427P230 | 2 per employee      |
| Red Warning Tag                        | 46-194427P322 | 2 per employee      |
| Safety Glasses                         |               | 1 per employee      |
| Safety Shoes                           |               | 1 pair per employee |
| Cut Resistant Gloves                   |               | 1 pair per employee |
| Non-Ferrous Flat Head Screwdriver      |               |                     |
| Non-Ferrous Knife                      |               |                     |
| Non-Ferrous Measurement Tape           |               |                     |
| Scissors                               |               |                     |

## 1.5 Field Supplied Materials

| Description | Quantity |
|-------------|----------|
| Tape        | 1 roll   |
| Tie-Wraps   | 2-8      |

## 1.6 Summary of Tasks

- Section 2 - Install Bore Liner and Replace Bridge Hardware.
- Section 3 - Install Chilled Air Blower (CAB) unit.
- Section 4 - Install Pen Panel Waveguide if necessary.
- Section 5 - Connect air hose from Bore Liner through the pen panel to the CAB.
- Section 6 - Chilled Air Blower configuration setup.
- Section 7 – Load Option Key and test.
- Section 8 – Troubleshooting.
- Section 9 – Field Replaceable Units (FRU's).
- Appendix A – PDU LOTO procedure.

## 1.7 Safety



**⚠ DANGER**

**LIKELY FATAL PERSONAL INJURY!!**

**LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN WHEN ALL PDU BREAKERS ARE OFF.**

**CHECK THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.**

## 2 Install Bore Liner and Replace Bridge Hardware

### 2.1 Prepare System for Lock-Out Tag out.

- 2.1.1 With the patient transport docked, move the carriage to the rear of the magnet by pressing the "In Fast" button on the enclosure.

### 2.2 Perform PDU LOCKOUT TAGOUT OF THE SYSTEM (15 MINUTES)

See Appendix A for Details.



**POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. PERFORM LOTO BEFORE INSTALLING THIS OPTION.**

### 2.3 Bridge Removal and Bore Liner Installation

- 2.3.1 Remove Front Split Bridge:

#### **CAUTION**



**FERROUS MATERIAL HAZARD!!**

**THE FRONT PULLY ASSEMBLY MAY CONTAIN PARTS MADE WITH FERROUS COMPONENTS. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL FORCIBLY ATTRACT IT TO THE MAGNET.**

**TO PREVENT POSSIBLE BODILY INJURY, OR DAMAGE TO COMPONENTS, PUT THE ASSEMBLY IN PLACE DIRECTLY, NOT MOVING IT IN THE MAGNETIC FIELD MORE THAN IS NECESSARY.**

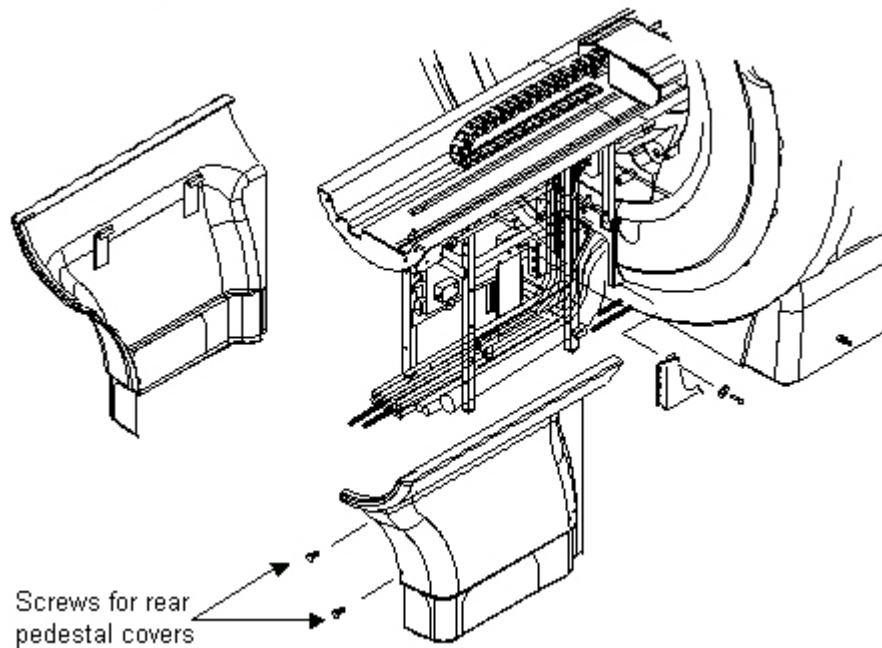
- Undock and move patient transport out of the way of the front cover of the magnet.
- Remove the two rear pedestal magnet corner covers. See Illustration 2.

**Illustration 2: MAGNET CORNER COVERS**



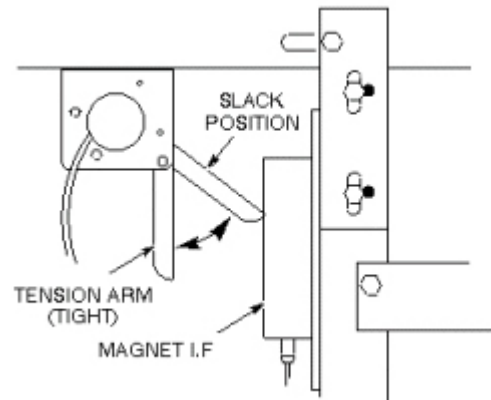
- c. Remove the rear pedestal covers. See Illustration 3.

**Illustration 3: REAR COVERS**



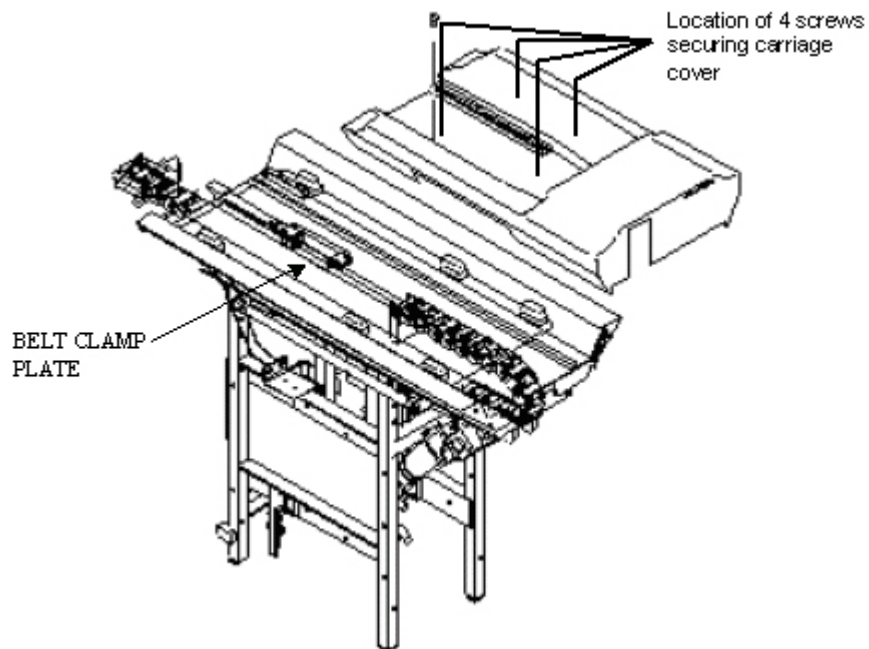
- d. Remove four screws from the top of the carriage cover. See Illustration 5.
- e. Lift the carriage cover from the carriage.
- f. Lift the tension arm to release tension from the carriage drive belt. See Illustration 4.

**Illustration 4:** RIGHT SIDE VIEW OF REAR PEDESTAL (AS VIEWED FROM REAR)



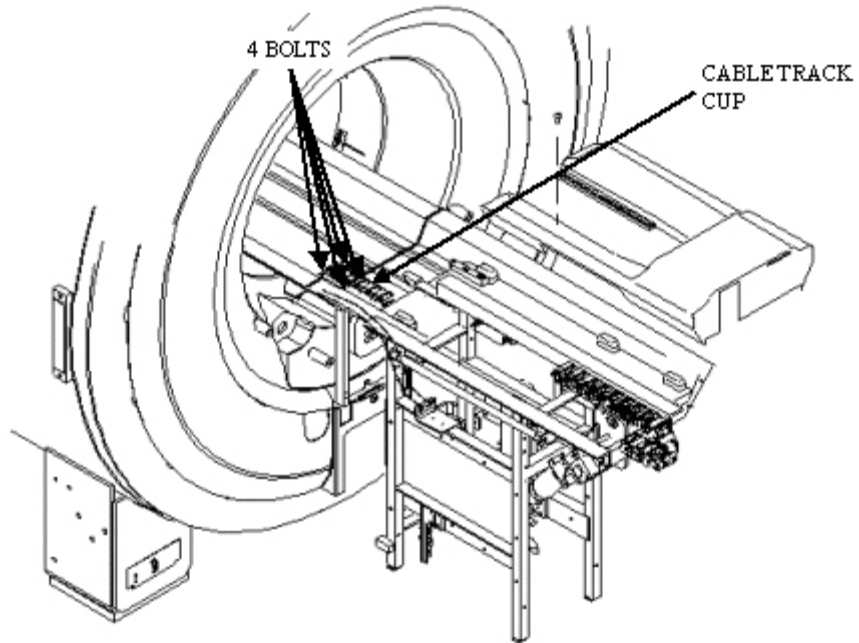
- g. Unclamp both ends of the belt by removing the four screws holding down the two belt clamp plates. See Illustration 5.

**Illustration 5:** BELT CLAMP PLATES



- h. Remove the rear section of the belt from being routed through the drive motor and rear pedestal. DO NOT pull the belt completely out from beneath the bridge.
- i. Remove the four bolts holding the cable track cup. (only two bolts on a HD or HDx system) See Illustration 6.

**Illustration 6: CABLE TRACK PLATE**

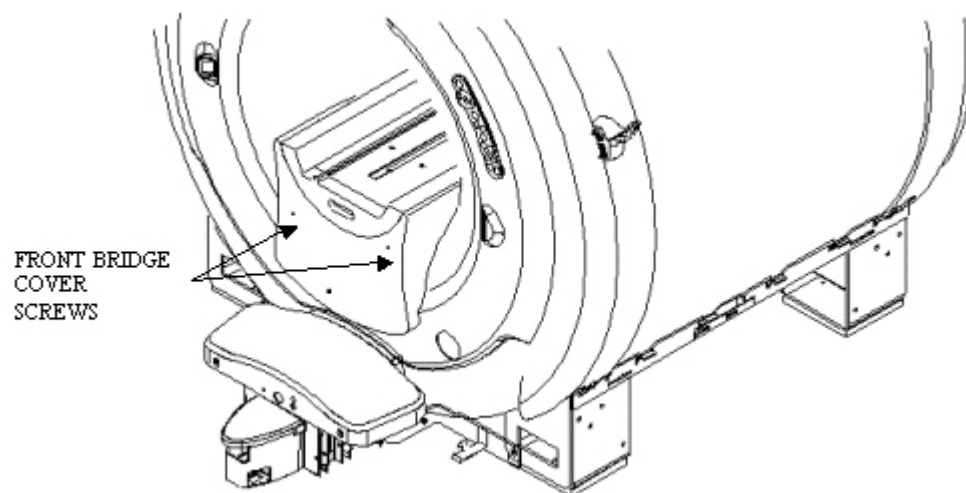


- j. Remove the four M10 hex nuts holding the front bridge to the rear bridge. The nuts are located on the rear pedestal side of the mating flanges.
- k. **For G3 systems only**, remove the two screws holding the rear bridge to the rear pedestal and lift off the rear bridge.

**NOTE:** The carriage cover cables do not need to be disconnected from the rear pedestal.

- l. Remove the two screws holding the front bridge cover. See Illustration 7.

**Illustration 7: FRONT BRIDGE CONNECTION**





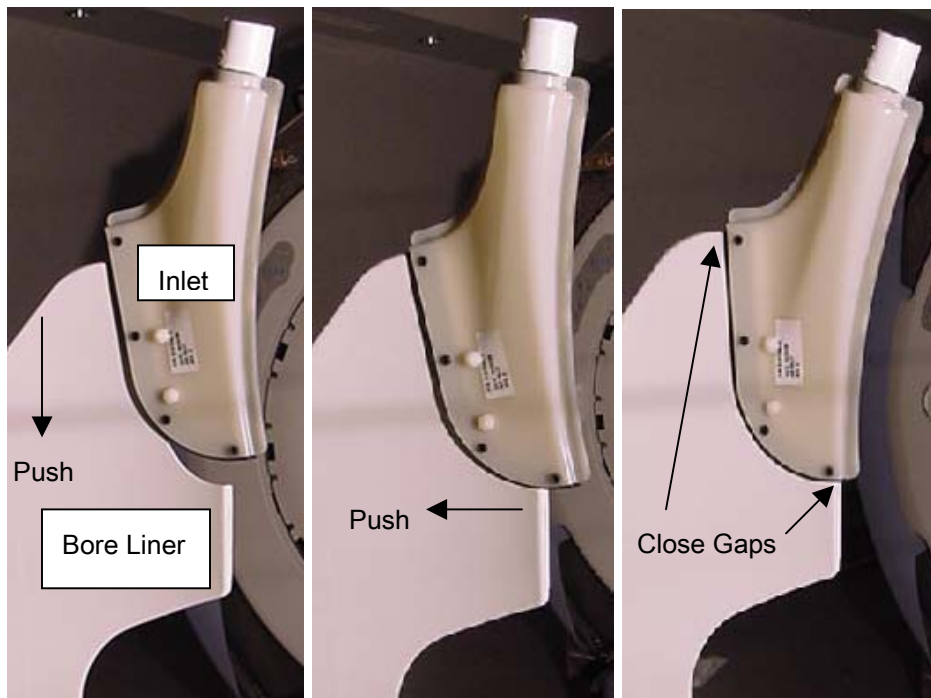
- m. Remove the two bolts that attach the bridge bracket to the magnet end ring.
- n. Carefully lift and slide the existing front bridge out of the magnet bore.

2.3.2 Remove the cosmetic strap that covers the gasket between the rear endbell and the RF Body Coil.

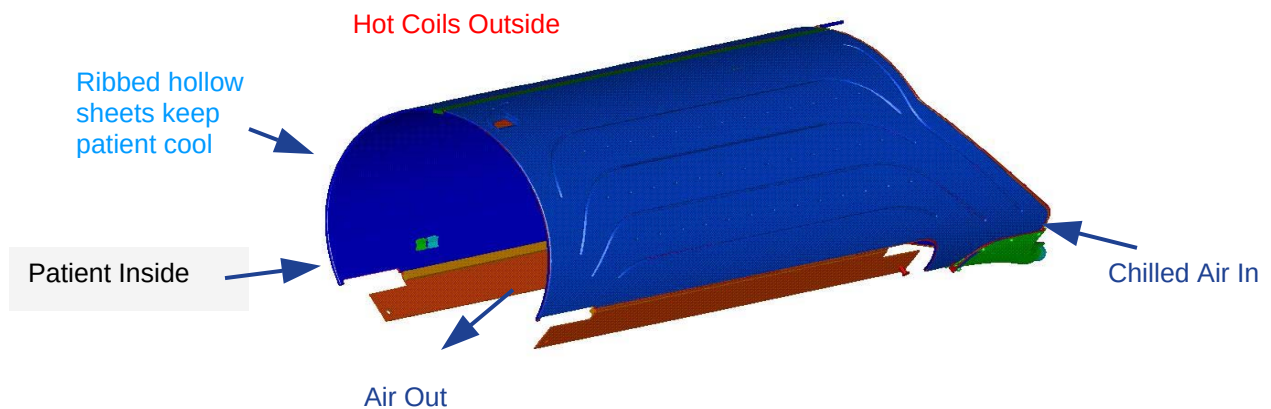
2.3.3 Bore liner preparation

- a. Attach right inlet to right bore liner and left inlet to left bore liner. See Illustration 8a. Be sure the connection is as tight as possible.

**Illustration 8a: INLET AND BORE LINER**

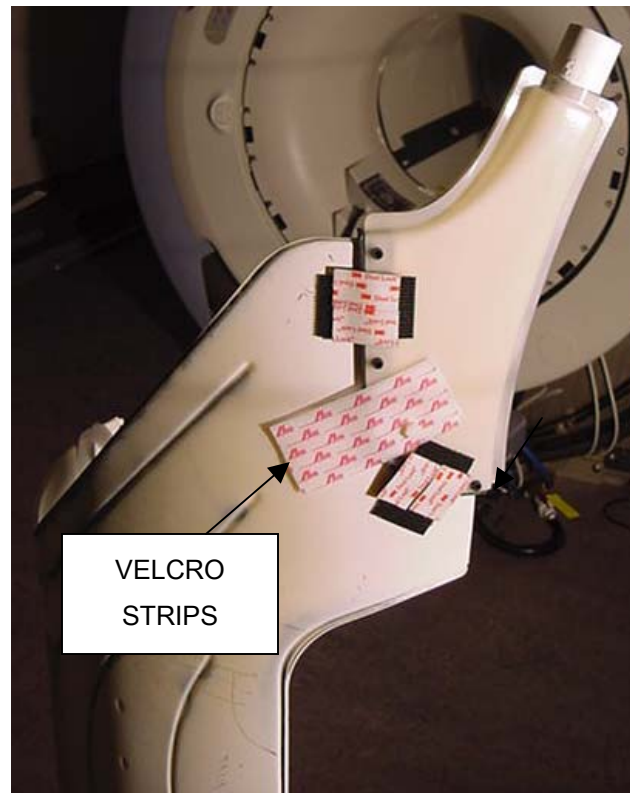


**Illustration 8b: BORE LINER WITH INLETS ATTACHED**



- b. Cut two of the four 6" (150 mm) dual lock pieces (Velcro straps) into 3" (75 mm) pieces. Attach the 3" pieces onto the bore liners and the inlets. Attach to convex side, not concave side. Use the 6" pieces to connect the two. See arrows in Illustration 9 for location.
- c. Attach a 1.5" (38 mm) PVC hose to each inlet. If necessary, use tie-wraps to attach the hose to each inlet. Avoid using ring clamps, to prevent possible white pixel problems.

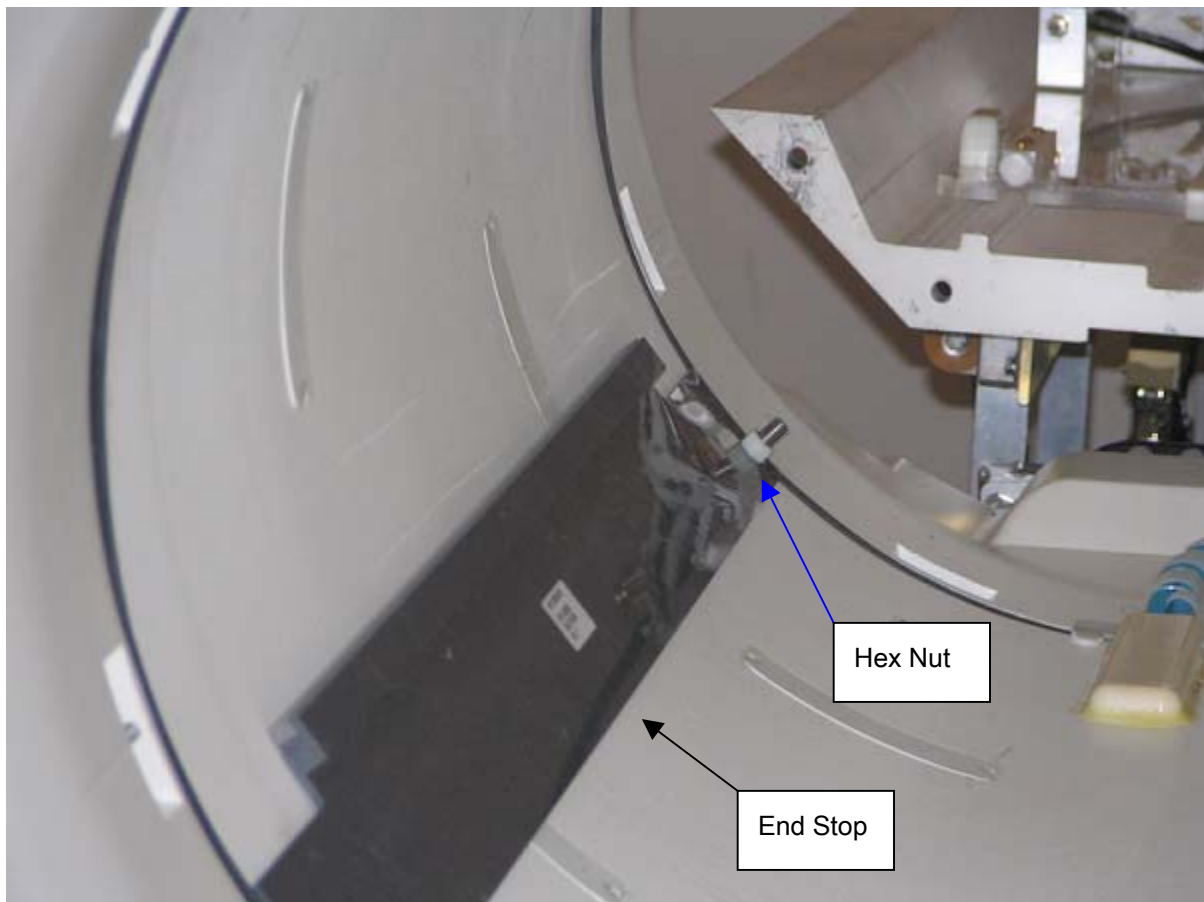
**Illustration 9:** ATTACH BORE LINER TO INLET WITH VELCRO STRIPS



#### 2.3.4 Bore Liner Installation

- a. Remove the Patient Comfort Module.
- b. Remove the bore light connections at the front of the bore and pull the bore lights to the rear of the bore. The bore lights will be removed later.
- c. Remove the plastic covers between the front edge of the RF Body coil and the front end bell.
- d. Tape the bore temperature sensor to the groove in the outer endbell.
- e. Attach left and right end stops with four hex nuts. See Illustration 10.

**Illustration 10: END STOPS**



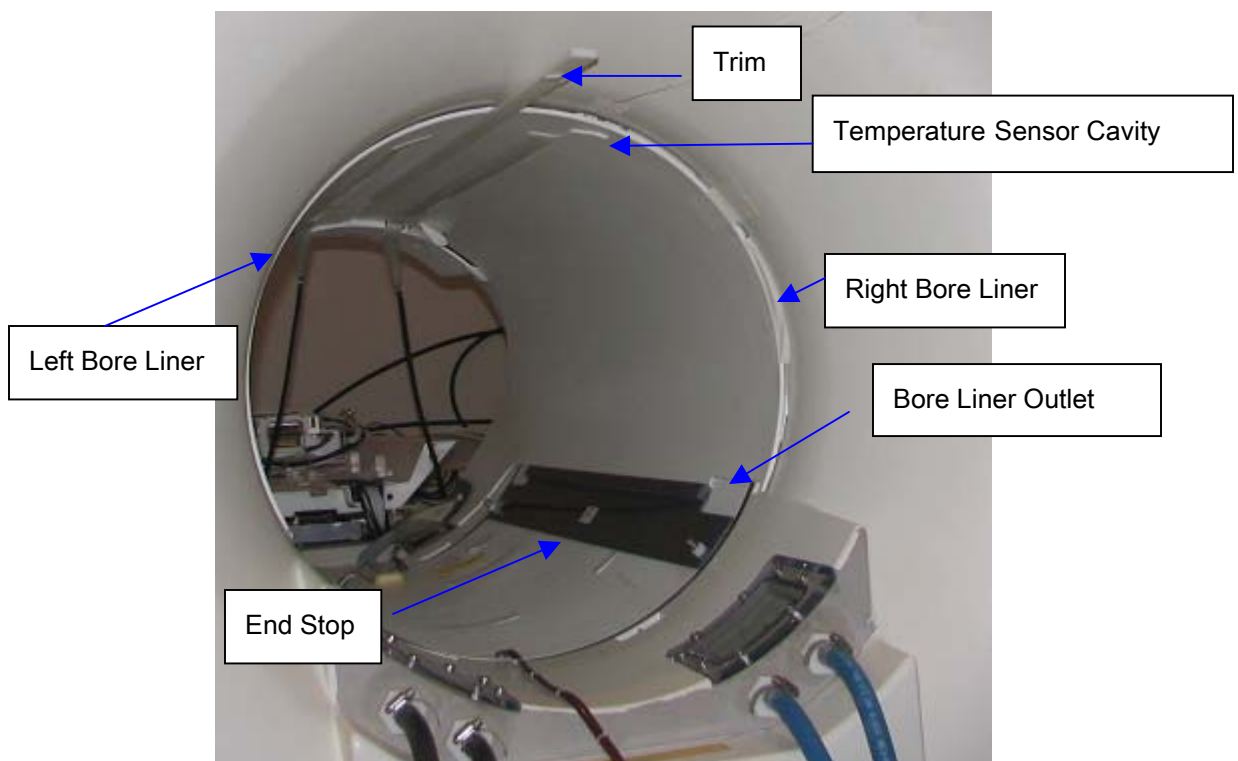
- f. Position the bore liners inside the bore, on the end stops, with inlets under the rear pedestal, and the PVC hoses snaked out the back of the rear pedestal. Bore liner outlet edge should line up with edge of end stop. This will ensure that the cavity on the right bore liner matches with the bore temperature sensor. See Illustrations 11a and 11b for details.

**NOTE:** If there is an uneven gap between the bore liner and the RF Tube and the bore liner cavity is over the bore temperature sensor, extra foam tape is included in the kit that can be applied to the back of the bore liner to reduce the gap.

**Illustration 11a:** BORE LINER OUTLET EDGE



**Illustration 11b:** BORE LINER ON END STOP



- g. Insert the trim piece. See Illustration 11b. The thick edge of trim piece should be on the bottom. Use 2 nylon setscrews to hold the trim piece to the bore at the rear. 2 extra setscrews are included.

## 2.4 Fiber optic routing

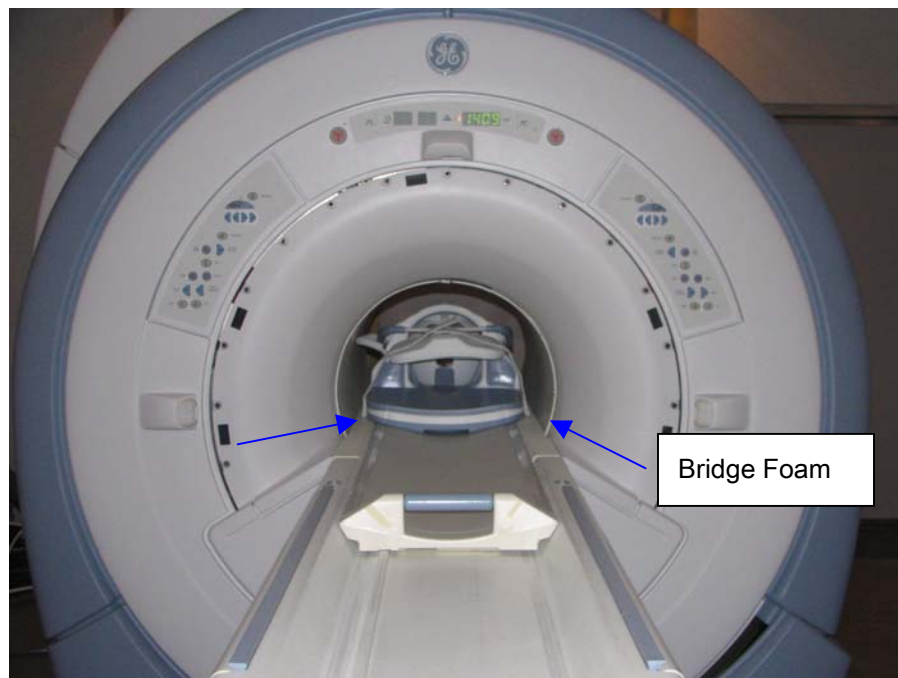
- 2.4.1 Remove the old bore lights. Check location of fiber optic control board. (usually right rear)
  - a. Loosen rear arc cover on same side as fiber optic control board.
  - b. Bring out rear arc cover about  $\frac{1}{4}$  inch, or enough to allow cable routing.
- 2.4.2 Route the new fiber optic harness through patient comfort module, around the bore, and plug into fiber optic control board.
- 2.4.3 Replace rear arc cover and patient comfort module.

## 2.5 Bridge preparation and installation

- 2.5.1 Attach the blue foam to each side of the new front bridge. Begin at the front edge and continue towards the back to fill the space between the bore liner and the bridge edge. See Illustration 12 for the location.
- 2.5.2 Remove the front bracket and belt off the old front bridge and transfer to the new one.
- 2.5.3 **For G3 systems only**, attach the split bridge chain bracket to the rear bridge.
- 2.5.4 Install the new bridge hardware. Reverse the instructions from 2.3.1 steps d – n.

**NOTE:** Make sure the front edge of the bore liner contacts the bridge to prevent a pinch point. The bore liner can be slid forward slightly if necessary to close any gaps larger than 9 mm. See Illustration 12.

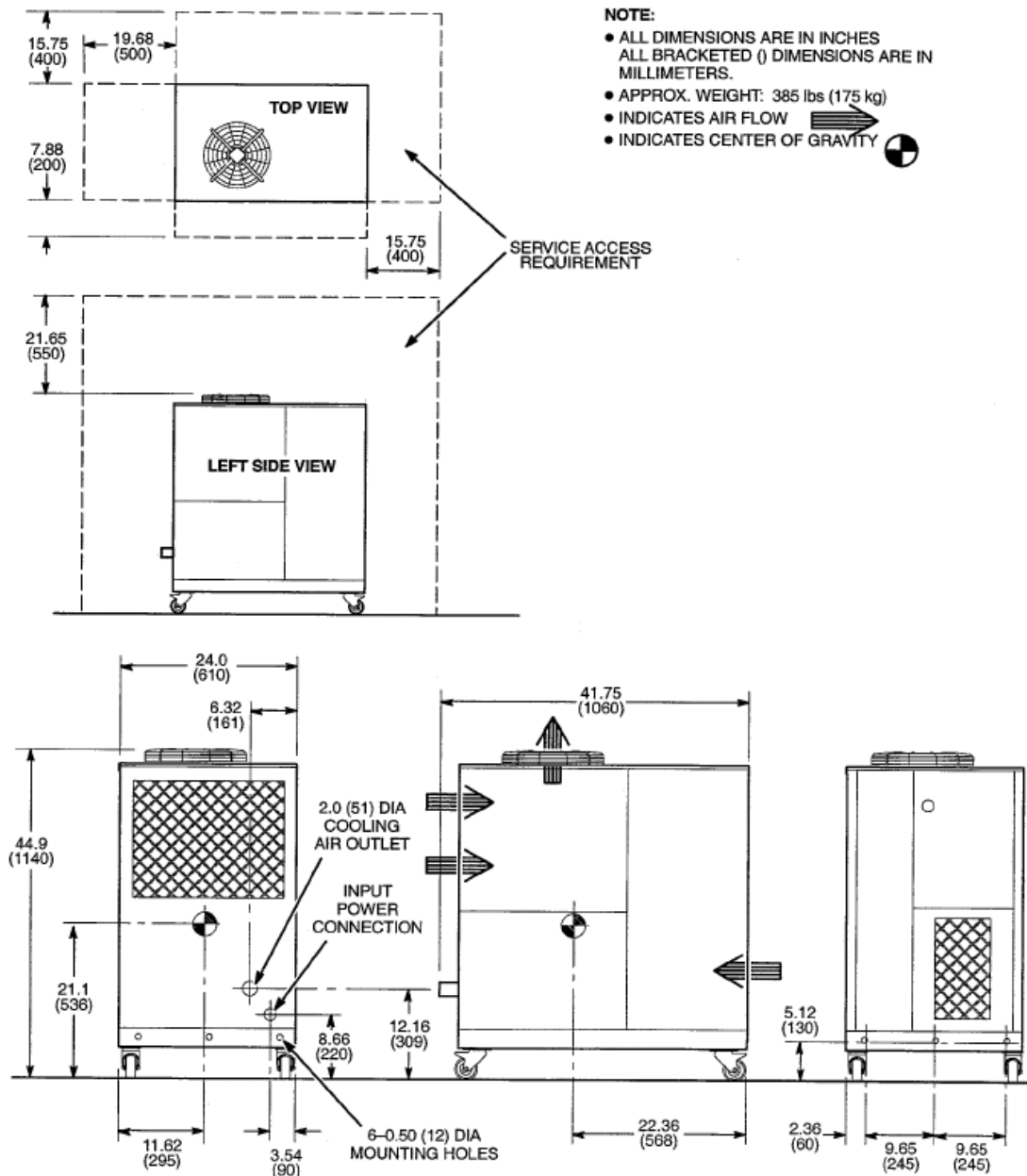
**Illustration 12:** NEW FRONT BRIDGE AND FOAM



### 3 Place Chilled Air Blower (CAB)

- 3.1 Move Chilled Air Blower into the equipment room making sure to allow the recommended clearances in front and behind the unit.
- NOTE:** The unit can be placed between two cabinets or in a corner as long as the front and rear service clearances are followed. (See Illustration 13 for the recommended service clearances.)

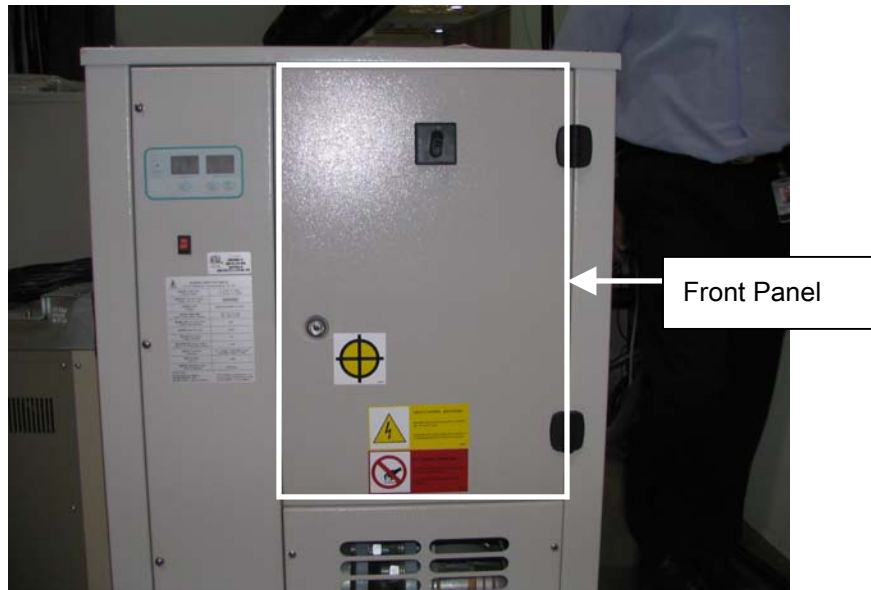
Illustration 13: CAB Service Requirements



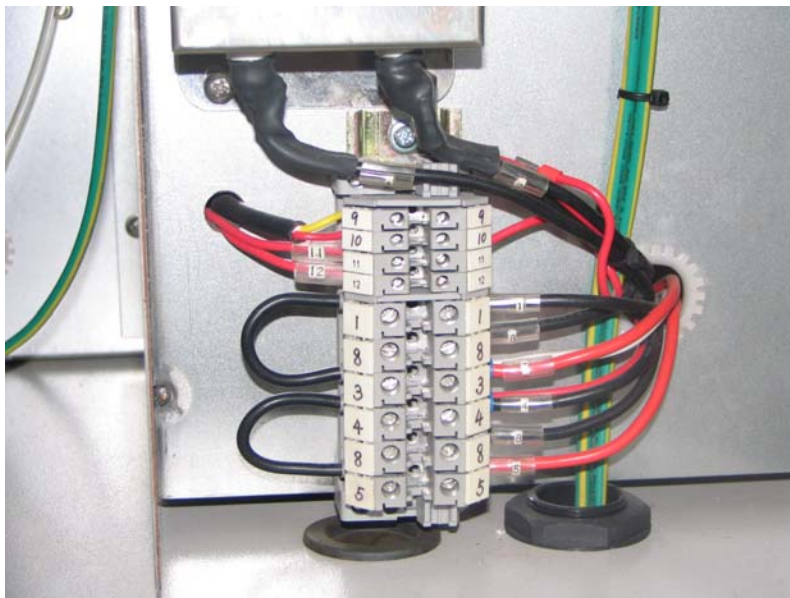


**3.2 Open the front panel and check the incoming power supply jumper settings on the Chilled Air Blower. See Illustrations 14a, 14b, and 14c.**

**Illustration 14a: CAB FRONT VIEW**



**Illustration 14b: CAB JUMPER TERMINAL BOARD**

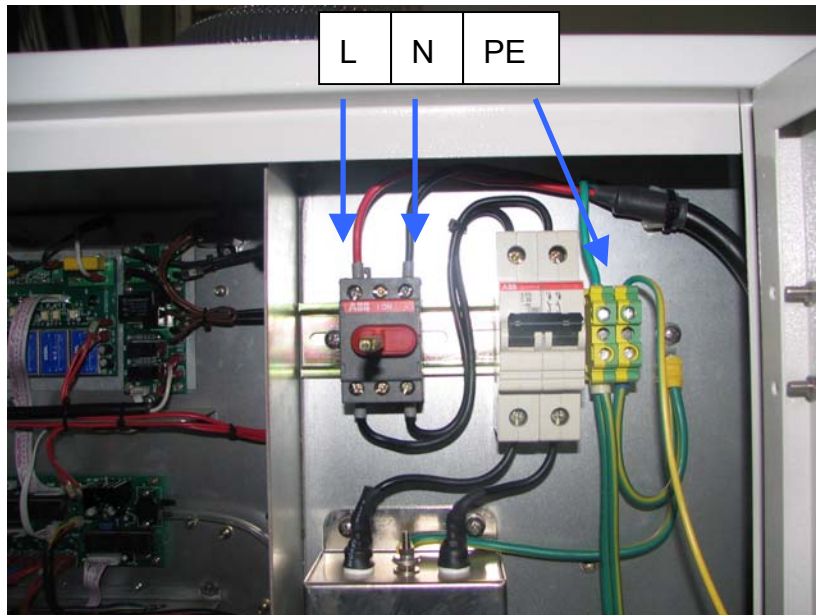


**Illustration 14c: CAB JUMPER SETTINGS**

| POWER SUPPLY |              | POWER SUPPLY SELECT<br>TERMINAL BLOCK CONNECT |   |   |   |   |   |
|--------------|--------------|-----------------------------------------------|---|---|---|---|---|
| RATED        | 277VAC       | 1                                             | 8 | 3 | 4 | 8 | 5 |
| RATED        | 200 ~ 240VAC | 1                                             | 8 | 3 | 4 | 8 | 5 |

### 3.3 Connect the CAB-MDP power cable to the input terminals on the Chilled Air Blower. See Illustration 15.

**Illustration 15: CAB POWER CONNECTIONS**



L= Line Voltage connection  
 N= Neutral Voltage connection  
 PE = Ground Wire connection

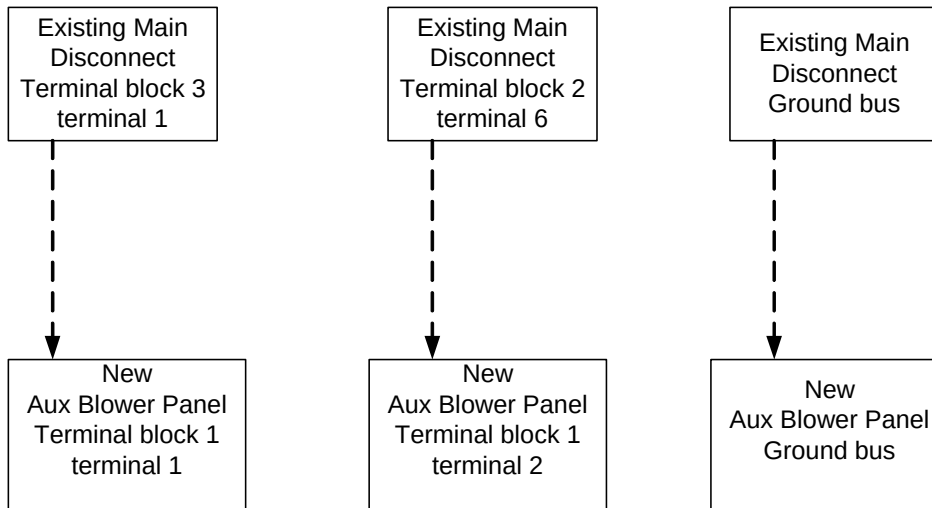


### 3.4 Confirm the CAB-MDP auto restart control wiring matches Illustration 16.

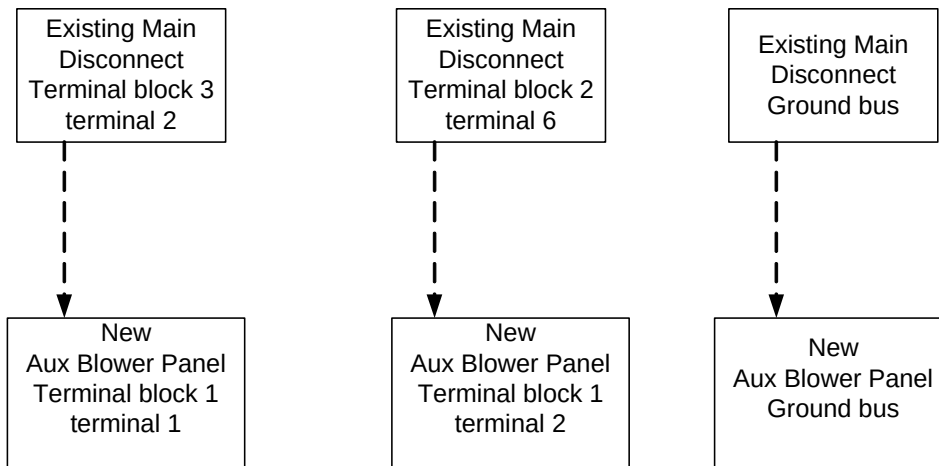
**Illustration 16:** MDP TO CAB-MDP CONTROL WIRING

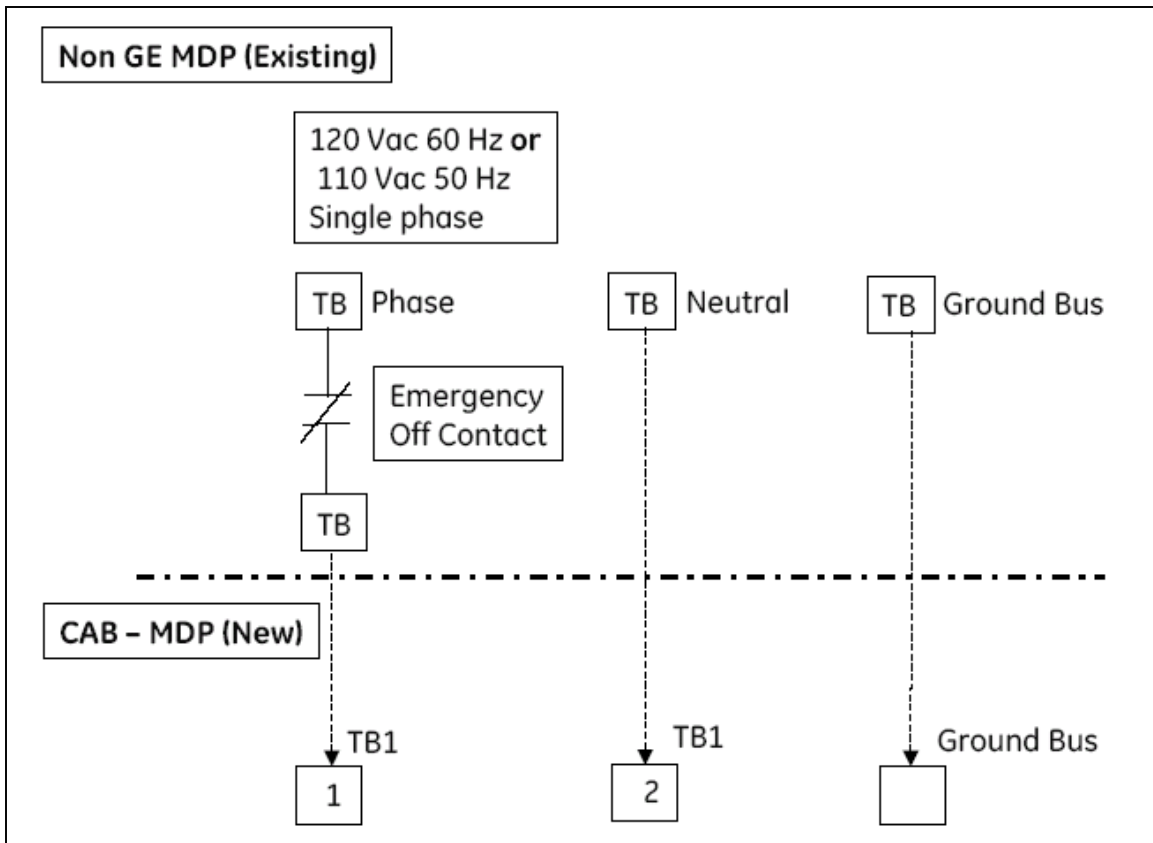
#### **CONTROL POWER CONNECTIONS IN EXISTING MAIN DISCONNECTS**

**GE #M3088TM also marked CR243B10453 wiring diagram 55-788127**



**GE #M3088TH also marked CR243B10435 wiring diagram 55-216870**





## 4 Pen Panel Waveguide Installation

### 4.1 Install the Pen panel modification (necessary if no waveguide available for the 2 ½" IPCM hose)

- 4.1.1 Remove the existing pen panel plate or pen panel waveguide. Remove any existing hose or cables if necessary. See Illustration 17

**Illustration 17:** EXAMPLE OF OLD PEN PANEL WAVEGUIDE



- 4.1.2 Tightly install the new 2 ½" waveguide / pen panel plate. See Illustration 18

**Illustration 18:** NEW PEN PANEL WAVEGUIDE



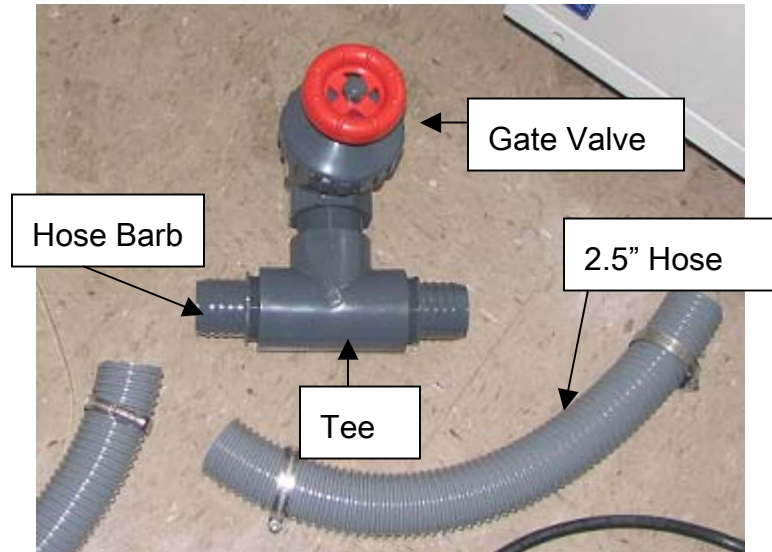
- 4.1.3 Re-route the existing hose/s through the waveguide and reconnect any removed cables.

**NOTE:** Wires and copper cables should not be passed through the waveguide into the Magnet Room or else image quality could be adversely affected.

## 5 Connect Hose from Bore Liner to Chilled Air Blower

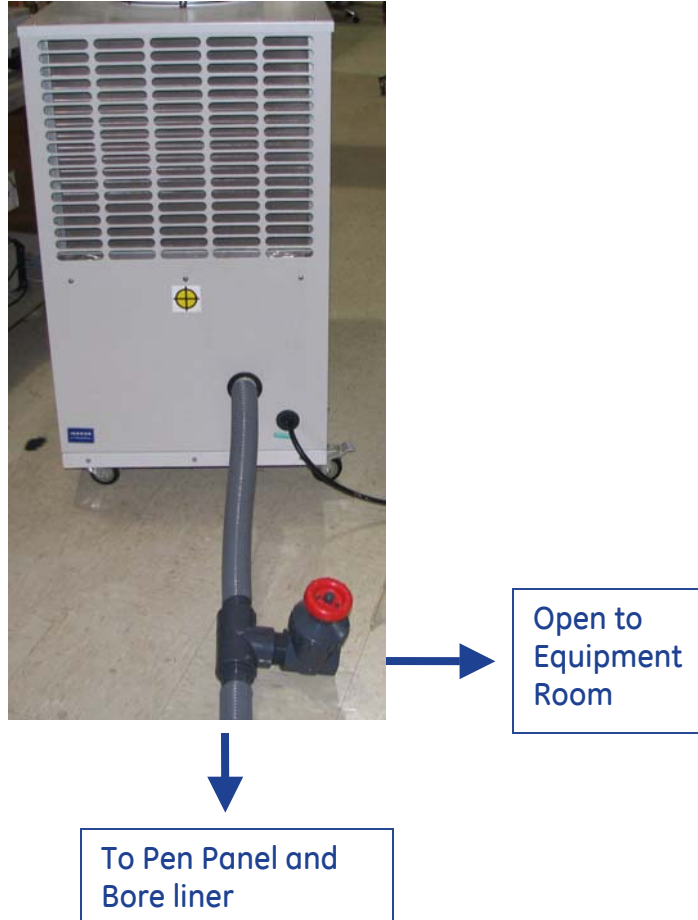
- 5.1 Assemble the 2" gate valve with the tee and barbed fittings. This valve will be used to adjust the airflow rate through the bore liners by relieving excess pressure in the line. The gate valve should be on the middle of the tee and the barbed fittings on the ends. See Illustration 19.

**Illustration 19:** GATE VALVE AND TEE CONNECTOR



**5.1.2 Confirm the gate valve is assembled as shown in Illustration 20.**

**Illustration 20:** GATE VALVE AND TEE CONNECTOR



**5.1.3 Connect the 1.5 inch hoses to the Y-pipe. Use wire ties or ring clamps if necessary. See Illustration 21.**

**NOTE:** The Y pipe is too large to fit inside the rear pedestal and will need to be placed outside of the rear pedestal. Route the 1.5 inch hoses from the air inlets through the rear pedestal to the Y pipe behind the rear pedestal.

**Illustration 21: Y-PIPE AND 1.5 INCH HOSES**



**5.1.4 Attach the 2 inch hose to other end of Y-pipe and push the remaining hose through the waveguide to the chiller.**

**5.1.5 Cut the 2 inch hose to length leaving the necessary slack at the chiller. Cut an additional 2 feet at the end of the 2 inch hose to allow for insertion of the gate valve.**

**5.1.6 Insert the gate valve between the Y-pipe and the Chilled Air Blower. See Illustration 20. Use ring clamps if necessary.**

**5.1.7 Remove the PDU LOCKOUT TAGOUT OF THE SYSTEM (15 Minutes)  
See Appendix A for Details**

## 6 Chilled Air Blower Configuration Setup

6.1 Plug in the CAB-MDP power cable into the CAB-MDP. This requires that the CAB-MDP is mounted within 15 feet (4.5 meters) of the CAB. See Illustration 22.

Illustration 22: CAB-MDP

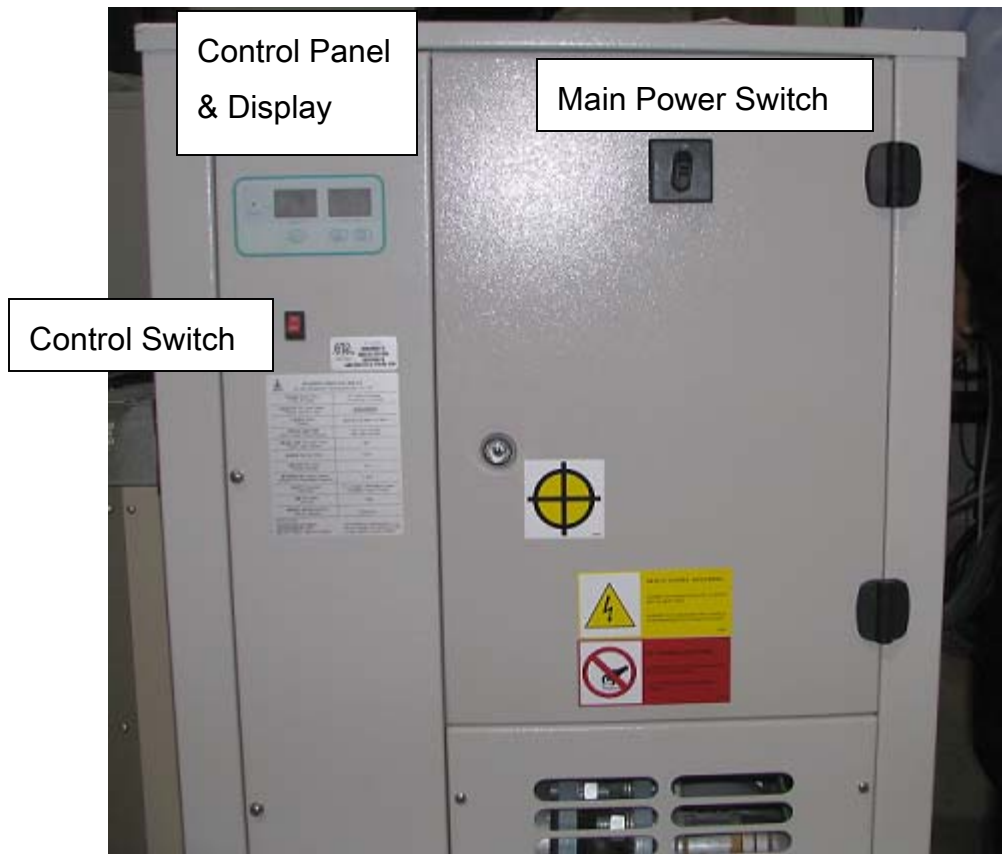


### 6.2 Turn on Chilled Air Blower

- 6.2.1 Switch on the CAB-MDP disconnect switch.
- 6.2.2 Turn on the CAB-MDP control switch. (Control Switch Light should turn on)
- 6.2.3 Turn on the CAB main power switch. See Illustration 23
- 6.2.4 Switch on the CAB control switch. See Illustration 23



**Illustration 23:** FRONT VIEW - CHILLED AIR BLOWER



### 6.3 Confirm the CAB Temperature Settings

- 6.3.1 The low temperature setpoint "A-1" should equal 10 C.  
a. The setpoints are entered through the control panel which is shown in Illustration 23. Press the "mode" button to display the various pages. The low temperature setpoint is displayed when the "A-1" page is shown. Press the up or down key to adjust if necessary. (see the Medicoool Blower 4.0 R407C manual for details)
- 6.3.2 The high temperature setpoint "A-2" should equal 20 C.  
a. The setpoints are entered through the control panel which is shown in Illustration 23. Press the "mode" button to display the various pages. The high temperature setpoint is displayed when the "A-2" page is shown. Press the up or down key to adjust if necessary. (see the Medicoool Blower 4.0 R407C manual for details)

### 6.4 Adjust the CAB pressure output

- 6.4.1 View the CAB output pressure on the control panel.  
a. The pressure feedback can be displayed on the control panel which is shown in Illustration 23. Press the "mode" button to display the various pages. The CAB output pressure is displayed when the "PrE" page is shown. (see the Medicoool Blower 4.0 R407C manual for details)
- 6.4.2 Adjust the Gate Valve until the CAB pressure output is 1.2 kPa.
- 6.4.3 Pull the two 1.5 inch ID hoses off of the Y-Pipe and check that the air flow is relatively balanced between the two bore liners. If air flow seems balanced, attach the two air hoses to the Y-Pipe.

## 7 Load Option Key and Test

### 7.1 Load the Software Option Key.

#### 7.1.1 Check Minimum Requirements:

**3.0T HD Systems: The minimum software level is 12.0 M5**

**3.0T HDx Systems: The minimum software level is 14.0 M3** (NOTE: if the system is a HDx upgrade site, the kit may contain the 12x IPCM option key. The 12x option key is upwardly compatible and can be used on a 14x system)

**NOTE: If your site doesn't contain the minimum software requirements, please contact the OLC.**

#### 7.1.2 Insert the Floppy.

#### 7.1.3 Select the [Common Service Desktop].

#### 7.1.4 Select [Configuration].

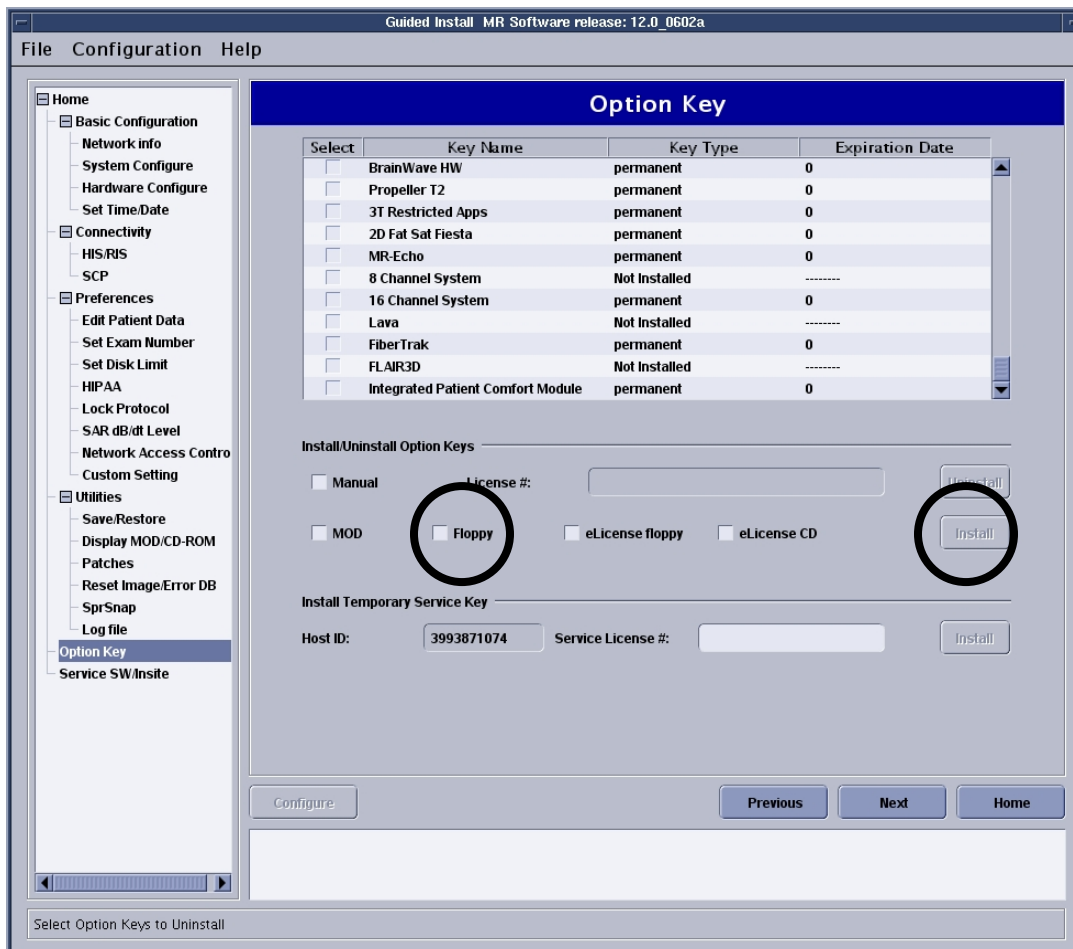
#### 7.1.5 Under Guided Install, select [FE mode]. Select [Click here to start this tool].

#### 7.1.6 At the password prompt, type operator and press [enter]

#### 7.1.7 Select "Option Key" Tab (see Illustration 24)

#### 7.1.8 Check the **Floppy** Box and then select [Install]. (See Illustration 24)

**Illustration 24: OPTION KEY**



- 7.2 Re-Install the rear pedestal covers.**
- 7.3 Re-Install the magnet corner covers.**
- 7.4 Perform test scan for both Head and Body coils to make sure system is working properly.**
- 7.5 Communicate the following information to the customer:**
  - 7.5.1 Provide the customer with the one page Chilled Air Blower Document for display near the operator desk. (The Chilled Air Blower Document is part number 5175579)
  - 7.5.2 Show the customer the location of the Chilled Air Blower (CAB)
  - 7.5.3 For HD and HDx systems, show the message that will appear on the monitor when an fMRI scan is started. See Illustration 25.

**Illustration 25: IPCM COZY BAR MESSAGE**



## 8 Troubleshooting

**8.1** This section contains common problems that could occur during the installation of the IPCM. See Table 3

**Table 3:** IPCM TROUBLESHOOTING

| Problem                                                                                     | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| There is too much air noise picked up by the microphone in the LPCA.                        | The CAB does increase the noise level slightly, but not enough to cause noise issues. (~2-3 db increase)<br>1) Check that the CAB output pressure is set to 1.2 kPa. (see section 6.4)<br>2) Check the bore liner inlets for air leaks.<br>3) Adjust the intercom board base amplification. See system intercom procedure.                                                                                                                                                                 |
| The bore liner temperature is too cold.                                                     | Adjust the lower temperature setpoint "A-1" on the CAB from 10 degrees Celsius to 15 degrees Celsius. (See section 6.3.1)                                                                                                                                                                                                                                                                                                                                                                  |
| There is a gap between the bore liner and the RF Body Coil at the front of the magnet bore. | 1) The bore liner can have a gap near the 1 O'clock location if the bore liner cavity doesn't line up with the bore temperature sensor. Adjust the bore liner (usually towards the front of the magnet) to line up the bore liner cavity with the bore temperature sensor. (see section 2.3.5.f)<br>2) If there is still an excessive gap between the bore liner and RF Body Coil, additional two sided tape has been included with the IPCM kit for filling the gaps. Apply as necessary. |
| Cannot set the output pressure to 1.2kPa.                                                   | Check the positioning of the gate valve. Make sure the gate valve relieves the excess pressure into the equipment room. (see Section 5.1.2)                                                                                                                                                                                                                                                                                                                                                |
| Cannot load the IPCM option key.                                                            | Check that the system has the minimum software requirements.<br>1) 3.0T HD minimum requirement - 12.0 M5<br>2) 3.0T HDx minimum requirement - 14.0 M3                                                                                                                                                                                                                                                                                                                                      |

## 9. FIELD REPLACEABLE UNITS (FRU'S)

| Description                         | M3033LG<br>Part<br>Number | M3033LH<br>Part<br>Number | M3033LJ<br>Part<br>Number |
|-------------------------------------|---------------------------|---------------------------|---------------------------|
| Bore Liner Hardware                 | 5126928                   | 5126928                   | 5126928                   |
| Chilled Air Blower (CAB)            | 5118421                   | 5118421                   | 5118421                   |
| Chilled Air Blower Air Filter       | 5118421-2                 | 5118421-2                 | 5118421-2                 |
| Thermal Bore Liner Assembly Parts   | 5132472                   | 5132472                   | 5132472-2                 |
| Auxiliary Power Panel (CAB-MDP)     | 5133079                   | 5133079                   | 5133079                   |
| G3 and HD Bridge with Notched Sides | 2386348-2                 | 2386348-2                 |                           |
| Rear Bridge                         | 2386347                   |                           |                           |
| Hdx Bridge with Notched Sides       |                           |                           | 5150910-2                 |
| 12x Software option key             |                           | 5165551                   |                           |
| 14x Software option key             |                           |                           | 5164641                   |

## APPENDIX A – LOTO DETAILS

### Lock Out / Tag Out Procedure

Name Of Equipment: Gradient (HFD or ACGD)/PDU Cabinet

Number Of Locks: Two (2) per employee

Title(s) of Employees Authorized to Perform LOTO: **Field Engineers**

Title(s) of Affected Employees and How to Notify: **Hospital Personnel, Verbal Communication.**

| ENERGY SOURCE        | YES | NO | LOCATION OF ENERGY ISOLATING MEANS    | MAGNITUDE OF ENERGY |
|----------------------|-----|----|---------------------------------------|---------------------|
| Electrical           | x   |    | PDU "Input" and "Service Disconnect"  | 208Vac, 420Vac      |
| Pneumatic            |     | x  |                                       |                     |
| Hydraulic            |     | x  |                                       |                     |
| Gas/Water/Steam      |     | x  |                                       |                     |
| Chemical             |     | x  |                                       |                     |
| Mechanical Motion    |     | x  |                                       |                     |
| Gravity              |     | x  |                                       |                     |
| Springs              |     | x  |                                       |                     |
| Thermal              |     | x  |                                       |                     |
| Stored Energy        | x   |    | Time discharge of internal capacitors | 208Vac, 420Vac      |
| Air Under Pressure   |     | x  |                                       |                     |
| Oil Under Pressure   |     | x  |                                       |                     |
| Water Under Pressure |     | x  |                                       |                     |
| Gas Under Pressure   |     | x  |                                       |                     |
| Steam                |     | x  |                                       |                     |
| Other                |     | x  |                                       |                     |

Type(s) of Equipment and/or Method(s) Selected to Dissipate or Isolate Stored Energy:

**Time method for dissipation of stored energy – 5 Minutes**

Type(s) of Equipment and/or Method(s) Used to Ensure Disconnection's:

**2 Personal locks and tags**

Name of Equipment: Gradient (HFD or ACGD)/PDU Cabinet

| SEQUENCE | LOCK OUT/TAG OUT PROCEDURE                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Prepare for shutdown of equipment. Notify affected personnel working in the area that LOTO is being performed.                                                                                                                                                                                                                                                                                                                                                                                      |
| 2        | Bring the PC software down. (not necessary on HDx systems)<br>a. Toggle mouse control to the PC monitor using the hardkey on the top, left side of the keyboard housing.<br>b. From the PC monitor use the mouse to select <b>Start► Shutdown► Shutdown►</b>                                                                                                                                                                                                                                        |
| 3        | Bring the Signa software down.<br>a. Toggle mouse control back to the Signa monitor using the hardkey on the top, left side of the keyboard.<br>b. Single-click on the Toolbelt icon.<br>c. Single-click on the System Shutdown softkey on the Service Desktop Manager.<br>d. Select OK to confirm the shutdown.<br>e. Wait for the system to indicate on the monitor that it is safe to power off the computer before proceeding. This usually takes about 90 seconds before this message is seen. |
| 4        | Remove the cover from the front of the gradient/PDU cabinet.                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5        | Power off the following circuit breakers:<br>a. Operators Workspace PC (not necessary on HDx systems)<br>b. Operators Workspace Host                                                                                                                                                                                                                                                                                                                                                                |
| 6        | Locate the main power supply (Service Disconnect) feeding the Gradient/ PDU Cabinet.                                                                                                                                                                                                                                                                                                                                                                                                                |
| 7        | Lock Out / Tag Out the Gradient/PDU Service Disconnect.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 8        | Rotate the larger main breaker labeled INPUT counter-clockwise to the green, O (OFF) position. This breaker is located on the front of the Gradient/PDU.                                                                                                                                                                                                                                                                                                                                            |
| 9        | Lock Out / Tag Out the PDU "INPUT" circuit breaker.<br>a. Press on the left-pointing arrow on the end of the handle to expose the location where the lock and tag can be placed.<br>b. Lock and tag out the PDU "INPUT" circuit breaker                                                                                                                                                                                                                                                             |
| 10       | Verify that all energy has been dissipated. Measure the power at the front panel 420 and 208 test points of the PDU.<br><b>Note:</b> the test points are 100 :1 ratio (208 Vac = 2.08 Vac 420Vac = 4.20Vac)                                                                                                                                                                                                                                                                                         |
| 11       | Service/repair equipment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 12       | Notify affected personnel that equipment will be re-energized and LOTO devices are being removed.                                                                                                                                                                                                                                                                                                                                                                                                   |
| 13       | Remove Lock-Out Tag-Out equipment from PDU and re-apply power to the gradient/PDU Cabinet.<br>a. Rotate the larger main breaker labeled INPUT clockwise to the Red, 1 (ON) position                                                                                                                                                                                                                                                                                                                 |

| SEQUENCE | LOCK OUT/TAG OUT PROCEDURE                                                                                                            |
|----------|---------------------------------------------------------------------------------------------------------------------------------------|
| 14       | Remove Lock-Out Tag-Out equipment from the main power supply (Service Disconnect)                                                     |
| 15       | Power on the appropriate circuit breakers.<br>a. Operators Workspace PC (not necessary on HDx systems)<br>b. Operators Workspace Host |
| 16       | Press "Reset EMO" button.                                                                                                             |
| 17       | Bring the Signa software back up.                                                                                                     |
| 18       | Bring the PC software back up. (not necessary on HDx systems)                                                                         |
|          |                                                                                                                                       |